

Roof Lifecycle Cost Analysis

Planned maintenance is projected to save the owner of a typical Hampton Roads mixed-use building **\$277K** over the next 20 years.

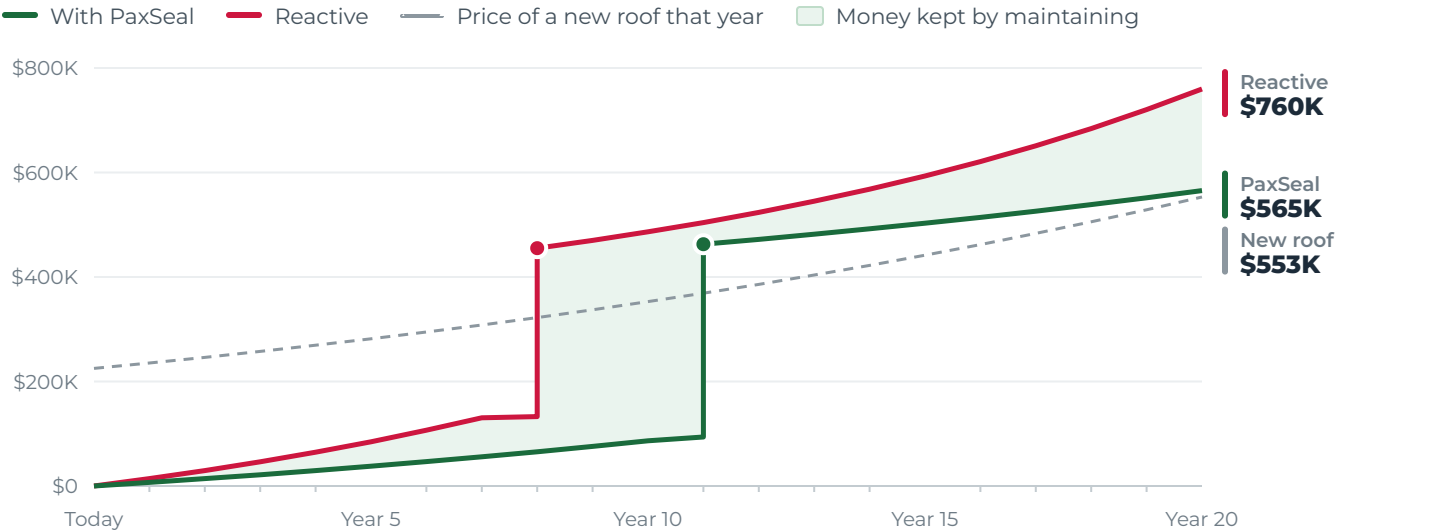
That is 32.0% less than fixing problems only after they leak. Under conservative to upside assumptions, the savings range from \$135K to \$367K. In today's dollars the estimate is worth \$145K.

MARKET	PROPERTY	ROOF AREA	SYSTEM	AGE	NEW ROOF TODAY
Hampton Roads	Mixed Use	25,000 sq ft	TPO	8 yrs Good	\$225K

PAXSEAL NET COST \$588K \$565K spent plus roof wear, net of roof life left	REACTIVE NET COST \$865K \$760K spent plus roof wear, net of roof life left	VALUE PROTECTED \$159K \$11K/yr more NOI at a 6.6% cap rate	NEW ROOF PRICE 2× in 15 yrs \$225K today, \$553K by year 20
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Cumulative roof spending, next 20 years

Nominal dollars. Vertical jumps are full roof replacements.



Roof life

When each scenario needs a new roof.



Where the savings come from

20-year totals

	PAXSEAL	REACTIVE	SAVINGS
PaxSeal program fees	\$109K	—	-\$109K
Repairs paid by owner	\$55K	\$388K	+\$333K
Expected storm damage	\$33K	\$49K	+\$16K
Roof replacements	\$369K	\$322K	-\$47K
Current roof value used	\$135K	\$135K	—
Credit for roof life left	-\$113K	-\$30K	+\$83K
Net 20-year cost	\$588K	\$865K	+\$277K

What this means

- 1 **A later replacement.** The next new roof moves from year 8 to year 11, when it costs less in today's dollars.
- 2 **Smaller repairs.** Fixes made on scheduled visits avoid emergency work at about 3.6× the cost in a mixed-use building, saving \$333K.
- 3 **Both income streams protected.** A leak can close the ground-floor retail and damage the offices or apartments above.
- 4 **The program pays for itself.** Each \$1 of fees avoids about \$3.54 of other roof costs.

Local risk profile

Norfolk City, VA (51710)

Hurricane / tropical-storm wind

~4.8 tropical storms or stronger within 50 nm of Norfolk per 20 years (NOAA HURDAT2 1975–2024). Expected storm damage to this roof over 20 years is **\$33K** if maintained and **\$49K** if neglected. A documented maintenance history also helps when an insurer questions a claim.

Hail	Low	
Strong wind	Moderate	
Tornado	Moderate	
Hurricane	Moderate	
Winter weather	Moderate	
Ice storm	Moderate	
FEMA National Risk Index rating		

Key assumptions

The basis for each figure in this analysis.

ASSUMPTION	VALUE	BASIS
Roof life	20 / 15 yrs	Maintained / reactive. Fannie Mae service-life tables; +5 yrs per NRC Canada and warranty terms
Roof cost growth	4.6%/yr	BLS Producer Price Index, nonresidential roofing contractors, 2007–2025
Replacement cost	\$9.00/sq ft	Hampton Roads contractor price ranges
Reactive repair cost	3× planned × 1.2	Industry estimates; no primary study found. ×1.2 for leak damage in a mixed-use building
Storm damage	0.44%/yr	Of roof cost, before insurance. FEMA National Risk Index expected building loss, roof-related hazards
PaxSeal fee	\$4,050/yr	PAX program estimate for this roof size. Your proposal will state the actual fee
Cap rate	6.6%	Blend of 30% retail, 35% office and 35% multi-family rates until local data is added

Estimates, not a quote. Projections from Hampton Roads market research and industry data, in nominal dollars. The PaxSeal fee is an estimate; your proposal states the actual price. Net cost includes wear on the current roof, less roof life left at year 20. Storm damage is before insurance. Property value is the yearly gain in net operating income divided by the cap rate, for income property where the owner pays for the roof.